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Positional comparison of countermovement jump performance in American collegiate football athletes

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Abstract

Purpose: The purpose of this investigation was to examine countermovement jump differences between positional groups in American football.

Methods: 85 Division I collegiate football players were assigned to one of three groups (lineman = 32, midskill = 29, skill = 24). Each participant performed three CMJ trials using a portable force platform. A one-way analysis of variance was performed on all variables of interest. An a priori significance level of $p < 0.05$ was used for all analysis.

Results: Significant differences were present across all three groups for jump height ($p < 0.01$) and RSImod ($p < 0.001$) with the skill group having the highest values and the linemen having the lowest. Inversely, take-off momentum and body mass were the greatest in the linemen group with the lowest being the skill group ($p < 0.001$). Linemen had longer duration in the propulsive phase ($p < 0.01$), braking phase ($p < 0.01$), and time to take off ($p < 0.01$).

Conclusions: The differences between groups can aid in the design of training programs targeting specific traits for each positional group. Additionally, differences in positional metrics of interest may exist when using the CMJ to monitor fatigue status over the course of a competitive season.

Keywords

Fatigue, ground reaction force, take-off momentum, talent identification

Introduction

The countermovement vertical jump (CMJ) has gained popularity as an assessment of neuromuscular status/capacity across various athletic populations. This is in large part due to the lack of additional fatigue that the assessment produces as well as the ease of implementation.[1,2] With the growth in the utilization of the CMJ to assess athletic populations, an emphasis has been placed on not just determining maximal performance but also gaining a better understanding of the underlying mechanics and strategies that are used to achieve a maximal jump height.[3,4] This has allowed for more targeted training interventions to improve overall athletic performance. Specifically, the use of the CMJ as an assessment tool in American football has become of interest because of the shared kinematic and kinetic qualities between the required skills within the sport itself such as blocking, tackling, and sprinting.[5]

As each of the playing positions within the game of football has different task-specific demands, identifying differences in movement strategies and force generation capacities is of critical importance not only in the design of effective training interventions but also during the longitudinal monitoring of an athlete's development. Previous investigations have identified that the positional groups within American football have differing levels of on-field demands as well as physical performance attributes.[5–9] It has been shown that on-field performance differences between positional groups exist not only in the total distance traveled and maximum sprint velocity, but also in the number and magnitude of accelerations.[6,9] As for the physical performance between positional groups, differences in absolute strength, sprint times, change of direction tasks, and vertical jump performance have been measured.[7] In a sample

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of junior college players, differences between positional groups were seen in vertical jump height between lineman and skill (wide receivers and defensive backs) positions.[7] Additionally, differences were seen in peak power between positional groups when using regression equations.[7] When examining differences in physical performance parameters between the level of competition in football athletes, differences have been seen in several variables including vertical jump height.[10,11] Specifically, differences in jump height between each positional group when comparing National Collegiate Athletic Association (NCAA) Division I and II collegiate athletes have been noted.[11] In a more detailed force-time analysis of CMJ performance across 4 positional groups in NCAA Division I collegiate football, differences were seen in 35 of the 47 variables assessed.[5] Additionally, this investigation demonstrated that many of the variables that can be obtained during the CMJ test while using force platforms have considerable overlap in the shared variance to explain jump performance.[5] This led to the recommendation that researchers and practitioners select metrics that provide high levels of reliability and applicability to the performance of the athlete based on the sport and scenario in which the testing is being administered.

As American football is classified as a collision-based field sport that requires multiple high-intensity movements, one's ability to generate the greatest momentum is critical as it would help to stop or drive an opponent backwards. Using take-off momentum (take-off velocity×body mass) could be an effective tool for monitoring an athlete's adaptation to training or fatigue status during the competitive season.[12] Previous investigations have shown a positive relationship between sprint momentum and body mass in American football athletes.[13] However, a negative relationship existed between sprint velocity and jump height to body mass.[13] This demonstrates that athletes with greater body mass are inhibited during sprinting and jumping tasks as it is more challenging to accelerate one's mass. Yet, the momentum generated by the athlete is potentially more of an important attribute for collision-based sports such as American football, rugby, and Australian Rules Football. McMahon et al.[12] found that take-off momentum during the CMJ task displayed large positive correlation values to sprint momentum at a variety of distances in rugby league athletes. As discussed previously, the CMJ is commonly implemented into a testing battery for its ease of administration and lack of additional fatigue that is induced. Using take-off momentum as a metric in the monitoring of fatigue in a collision-based sport such as American football could provide a more meaningful metric. While the use of take-off momentum as a metric used during the CMJ assessment is relatively limited at this time, positional differences in take-off momentum have been seen in rugby league athletes with the larger forwards having a higher take-off momentum though they had significantly lower jump heights.[14]

As mentioned previously, many investigations examining positional differences in football athletes have used the vertical jump task but are limited to jump height based on the use of a Vertec device or contact mat.[7,15] Thus, identifying differences not only in the jump performance but also in the underlying strategies and mechanics is important to better develop training programs to maximize on-field performance. Furthermore, comparisons using variables such as take-off momentum in the comparison of playing positions within American football could provide more appropriate benchmarking and profiling of athletes.

Thus, the purpose of this investigation was to evaluate differences between positional groups in American football using both traditional metrics from the force-time analysis such as jump height, force generation capacities, and phase duration as well as an alternative metric in take-off momentum. We hypothesized that the skill players will have higher jump heights and RSImod. We hypothesis that the lineman group will have greater take-off momentum than the other two positional groups.

Methods

To evaluate positional differences in CMJ performance across NCAA Division I collegiate American football athletes, a cross-sectional research design was used during the early summer training block. Participants had just completed a two-week active recovery period of training during the semester break. All testing was conducted during one session and was a part of the regular athlete-testing program that all athletes participated in as part of their sports participation.

Participants

85 NCAA Division I collegiate football athletes (age 20.18 ± 0.76 years, height 183.94 ± 5.63 cm, body mass 105.98 ± 24.68 kg) were tested at the beginning of the summer training period as a part of their athlete monitoring program. Participants were assigned to one of three groups lineman (offensive and defensive, $n = 32$), midskill (linebackers, running backs, and tight ends, $n = 29$), and skill (wide receivers and defensive backs, $n = 24$). All participants were cleared to partake in all team-related activities by the university's sports medicine team. Before testing, all subjects provided informed written consent as approved by the university's institutional review board.

Procedures

Participants completed a warmup consisting of jumping jacks, knee hugs, leg swings, body weight squats, and sub maximal jumps. Each participant performed three CMJ trials on a portable force platform (AMTI, Accupower, Watertown, MA, USA). All trials were sampled at

1000 Hz. Arm swing was restricted by placing a polyvinyl chloride dowel (<1.0 kg) across their upper back similar to a high bar squat position. Participants were instructed to use a self-selected countermovement depth and stance. Instructions were also given before each trial for participants to "jump as high as possible". Each trial began with 1 s of quiet standing to calculate body mass from the vertical ground reaction force (GRF). The force – time curve of each trial was visually inspected to ensure one second of quiet standing after each trial was collected. A minimum of 30 s of rest was provided between trials.

Data analysis

Raw vertical GRF data where exported and analyzed using a customized Excel spreadsheet (Microsoft, Redmond, WA, USA) modeled using the methods described by Chavda et al.[16] Specific CMJ phase definitions followed those suggested by McMahon et al.[17] In brief, the phases of interest for the current study were defined as the unweighting, braking, and propulsive phases. Movement onset was determined using previously established methodologies of finding the mean vertical force of the one second of quiet standing.[3,17,18] The the standard deviation of the vertical force during quiet standing was then calculated. After multiplying the SD by 5, this value was subtracted from the mean force of quiet standing (mean force of quiet standing – 5*SD of quiet standing). The first instance in which vertical force fell below the calculated value was then identified as movement onset. Unweighting starts at movement onset until the vertical GRF then surpasses the mean force of quiet standing. Braking was defined as the end of the unweighting phase until the instant in which the center of mass velocity reached zero. The propulsive phase was defined as the end of the braking phase until the point of take-off. Take-off was determined as the instance in which vertical force fell below a threshold equal to the mean plus 5 SDs of the flight phase vertical force.[19,20] The center of mass velocity was calculated by first calculating the center of mass acceleration for each sample by subtracting the calculated mean force during quiet standing from the vertical GRF data. Then, integration of acceleration data with respect to time using the trapezoidal rule, beginning 30 ms before movement initiation provided center of mass velocity.[21] Integration of center of mass velocity data with respect to time provided center of mass displacement. As for specific variable calculations, time to take-off was defined as the duration from movement initiation to the instant of take-off. Reactive strength index modified (RSImod) was calculated as jump height divided by time to take-off.[22] Take-off momentum was calculated as the product of body mass and take-off velocity.[12] Jump height was determined using the velocity at takeoff method.[23] All force variables are presented as net force (measured vertical GRF – body mass). Variables of interest for this investigation included phase specific peak and mean force and power, force at zero velocity, force and velocity at peak power, phase durations, as well as the aforementioned calculated variables. These variables all for comparisons to previous investigation but also provide information regarding the movement strategy which each positional group is using.

Statistical analysis

Data normality was tested using Sharpio-Wilk tests and all data was normally distributed. Mean data of the three trials were used in all statistical analyses. The reliability of the variables of interest was assessed using the coefficient of variation (CV) and interclass correlation coefficients (ICC). All reliability analyses were conducted using all three trials, and ICC values were calculated using a two-way random-effects, absolute-agreement model (ICC [2, 1]). High reliability was determined as CV $\leq$ 5% and an ICC of greater than $\geq$0.9. Acceptable reliability was deemed to occur with a CV between 5% and 10% and an ICC of between 0.9 and 0.8.[3,24,25] Comparisons across the three positional groups were made using a one-way analysis of variance for each variable of interest. Fisher's least significant difference comparisons were used in determining where statistical differences were present between groups. Significance for all tests was a priori set at $p < 0.05$. All statistical analyses were performed using SPSS (v28.0, SPSS Inc., Chicago, IL, USA). Eta squared effect sizes were calculated for each variable and interpreted as small = 0.01, moderate = 0.06, and large = 0.14.

Results

All variables demonstrated acceptable levels of reliability (Table 1). All data are reported as group means and displayed in Table 2. Positional differences were present in jump height (F 2,94 = 37.89, $p < 0.001$, $\eta^2 = 0.45$). The skill group displayed the highest jump (44.63 ± 0.07 cm) followed by midskill (39.74 ± 0.05 cm) and lineman (31.15 ± 0.07 cm) groups with significant differences between all three groups. Similarly, RSImod showed significant differences between all three groups (F 2,94 = 42.05, $p < 0.001$, $\eta^2 = 0.49$) with the highest value coming from the skill group (0.55 ± 0.12) followed by midskill (0.48 ± 0.10) and lastly the lineman (0.33 ± 0.07).

Peak propulsive force was statistically different between groups (F(2,94 = 13.88, $p < 0.001$, $\eta^2 = 0.23$). The lineman group displayed statistically greater peak propulsive force (1664.72 ± 203.62 N) than both the midskill (1487.84 ± 250.21 N) and skill (1385.45 ± 198.12 N) groups. No other differences were seen between groups. A statistically significant difference was present in mean propulsive force (F 2,94 = 7.67, $p < 0.001$, $\eta^2 = 0.14$). Lineman displayed the greatest mean propulsive force (1116.66 ± 102.62 N) which was statistically greater than both the midskill

Table 1. Interclass correlation coefficients (ICC) and coefficient of variation (CV) for CMJ metrics.

	ICC (95% CI)	CV (95% CI)
Peak Braking Force (N)	0.94 (0.92–0.97)	9.11 (7.96–10.27)
Mean Braking Force (N)	0.96 (0.93–0.98)	9.47 (8.23–10.70)
Force at Zero Velocity (N)	0.95 (0.93–0.97)	2.09 (1.66–2.53)
Peak Propulsive Force (N)	0.94 (0.91–0.97)	5.98 (4.97–6.99)
Mean Propulsive Force (N)	0.90 (0.83–0.94)	6.45 (3.93–8.96)
Peak Propulsive Power (W)	0.98 (0.97–0.99)	2.58 (2.16–2.99)
Force at Peak Power (N)	0.97 (0.95–0.98)	9.47 (8.23–10.70)
Velocity at Peak Power (m/s)	0.97 (0.96–0.99)	4.46 (3.55–5.78)
Braking Duration (ms)	0.95 (0.91–0.97)	8.61 (7.26–9.95)
Propulsive Duration (ms)	0.95 (0.93–0.97)	4.52 (3.52–5.51)
Time to Takeoff (ms)	0.93 (0.88–0.96)	6.65 (5.55–7.75)
Countermovement Depth (m)	0.98 (0.96–0.99)	8.68 (7.56–9.81)
Jump Height (m)	0.98 (0.97–0.99)	4.79 (3.93–5.65)
RSI Mod	0.98 (0.97–0.99)	3.81 (3.19–4.42)
Jump Momentum (kg*m/s)	0.99 (0.98–0.99)	1.51 (1.22–1.79)
Take-off Velocity (m/s)	0.99 (0.98–0.99)	1.51 (1.22–1.79)

(1029.55 ± 184.17 N) and skill (968.80 ± 166.27 N) with no other significant differences between groups. Propulsive peak power was statistically different between groups ($F(2,94 = 2.58$, $p < 0.001$, $\eta^2 = 0.32$) The lineman group displayed significantly greater propulsive peak power (6314.96 ± 596.70 W) than both the midskill (5510.65 ± 769.40 W) and skill (5277.38 ± 635.93 W) groups. No differences in propulsive peak power were seen between the midskill and skill groups. Force at propulsive peak power was significantly different between groups ($F(2,94 = 22.58$, $p < 0.001$, $\eta^2 = 0.33$). Statistically significant differences were seen between all groups with the highest force output being in the lineman (1377.83 ± 164.67 N), followed by the midskill (1173.79 ± 167.32 N), and lastly the skill group (1116.33 ± 132.30 N). Velocity at propulsive peak power was statistically different in all groups between groups ($F(2,94 = 29.96$, $p < 0.001$, $\eta^2 = 0.39$). The skill group had the greatest velocity at peak power (2.71 ± 0.20 m/s) followed by midskill (2.59 ± 0.15 m/s) and last the lineman (2.33 ± 0.24 m/s). Peak braking force ($p = 0.052$, $\eta^2 = 0.06$), mean braking force ($p = 0.38$, $\eta^2 = 0.02$), force at zero velocity ($p = 0.06$, $\eta^2 = 0.06$) and countermovement depth ($p = 0.95$, $\eta^2 < 0.01$) displayed no statistically significant differences between positional groups.

Take-off momentum displayed statistically significant differences between all groups ($F(2,94 = 59.05$, $p < 0.001$, $\eta^2 = 0.56$) with the lineman (347.20 ± 39.00 kg*m/s) having the greatest momentum, followed by midskill (283.91 ± 33.88 kg*m/s) and lastly the skill (258.73 ± 28.14 kg*m/s) (Figure 1). A similar trend was seen with body mass ($F(2,94 = 165.94$, $p < 0.001$, $\eta^2 = 0.78$) where statistically significant differences were seen between all groups. The lineman had the greatest body mass (135.29 ± 15.26 kg), followed by the midskill (98.92 ± 10.80 kg), and lastly the skill (84.43 ± 7.90 kg). Velocity at take-off was significantly different between groups ($F(2,94 = 29.03$, $p < 0.001$, $\eta^2 = 0.39$). The skill group had the highest take-off velocity (3.06 ± 0.21 m/s), followed by midskill (2.88 ± 0.26 m/s), and lastly the lineman (2.61 ± 0.21 m/s).

Discussion

The purpose of this study was to examine the positional differences in the CMJ tasks within a sample of NCAA Division I football athletes. The findings of this study are largely in agreement with previous literature exploring positional differences in American football.[5,7] Overall linemen appeared to have greater force-generating capacities during the propulsive phase while the skill positions used more of a velocity-driven movement pattern. This is demonstrated with similar countermovement depths between groups but differing propulsive phase durations. However, the current investigation also has conflicting evidence to previous investigations examining CMJ positional differences in American football athletes.[5]

This investigation demonstrates that different strategies are used between positional groups during the CMJ as peak braking force, mean braking force, and force at zero velocity and countermovement depth were not different between groups, while the propulsive phase demonstrated differences between groups. Variables used in assessing the strategy utilized during the CMJ typically include time to take-off, phase durations, and countermovement depth. Donahue et al.[3] demonstrated that female athletes across different sports displayed no differences in jump height, with differences in temporal variables and countermovement depth. This also coincided with differences in mean propulsive force, thus suggesting different strategies being used to achieve a similar jump height. In the present study, it is suggested that different strategies are used between positional groups, however this differs from previous investigations.[5,26] All groups used a similar countermovement depth in the current investigation. However, the lineman group had significantly greater durations across all temporal variables. Longer durations over the same countermovement depth created a slower movement velocity. The opposite occurred in the other groups which had shorter durations and thus higher movement velocities. The lineman group had to employ a different movement strategy than the other two groups due to their greater body mass which required more force. While different than previous investigations examining jump strategies, the current investigation demonstrates the need to understand positional differences within the same sport. Understanding the movement strategies that positional

Table 2. Countermovement jump comparison across positional groups (mean ± SD).

	Lineman	MidSkill	Skill	p	η^2
Peak Braking Force (N)	1313.68 ± 281.31	1320.98 ± 291.14	1168.89 ± 271.69	0.052	0.06
Mean Braking Force (N)	615.98 ± 188.55	674.03 ± 220.67	606.58 ± 204.17	0.382	0.02
Force at Zero Velocity (N)	1311.59 ± 283.56	1320.77 ± 291.10	1168.60 ± 271.82	0.055	0.06
Peak Propulsive Force (N)	1664.71 ± 203.61*†	1487.83 ± 250.21	1385.45 ± 198.11	<0.001	0.23
Mean Propulsive Force (N)	1116.65 ± 102.62*†	1029.55 ± 184.16	968.80 ± 166.26	<0.001	0.14
Peak Propulsive Power (W)	6314.96 ± 596.70*†	5510.65 ± 769.40†	5277.38 ± 635.93	<0.001	0.32
Force at Peak Power (N)	1377.83 ± 164.67*†	1173.79 ± 167.32†	1116.33 ± 132.30	<0.001	0.33
Velocity at Peak Power (m/s)	2.33 ± 0.24*†	2.59 ± 0.15†	2.71 ± 0.20	<0.001	0.39
Braking Duration (ms)	224.88 ± 45.68*†	188.59 ± 68.78	178.83 ± 50.35	0.003	0.12
Propulsive Duration (ms)	301.09 ± 35.06*†	266.92 ± 53.04	261.55 ± 35.69	<0.001	0.16
Time to Takeoff (ms)	970.94 ± 138.90*†	846.31 ± 161.23	845.37 ± 152.32	<0.001	0.14
Countermovement Depth (m)	0.33 ± 0.07	0.34 ± 0.049	0.34 ± 0.06	0.948	<0.01
Jump Height (m)	0.31 ± 0.07*†	0.39 ± 0.05 †	0.44 ± 0.06	<0.001	0.45
RSI Mod	0.32 ± 0.06*†	0.48 ± 0.09 †	0.55 ± 0.12	<0.001	0.48

*Significantly different from MidSkill.
†Significantly different from Skill.
ǀSignificantly different from Lineman.

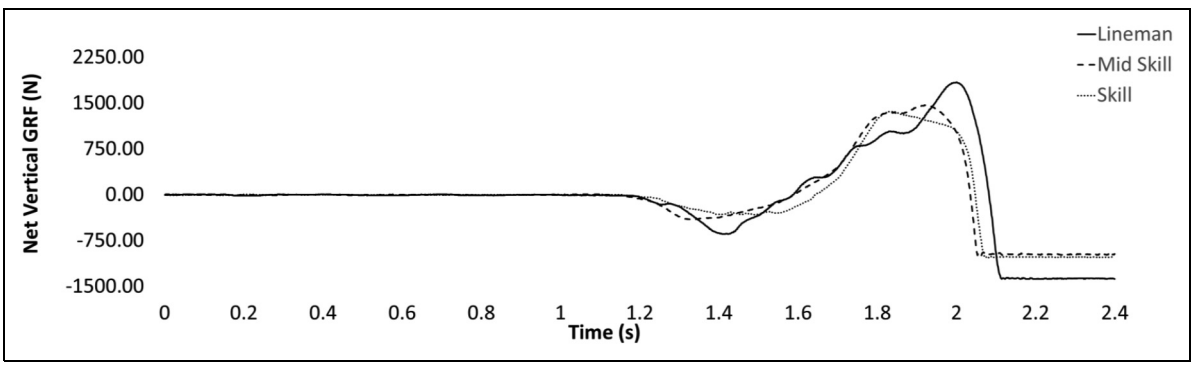

Figure 1. Representative force-time curve of each positional group.

groups use during specific tasks can not only help in identifying targeted metrics for training but also help in talent identification to put athletes in a position where they potentially best fit from a physical performance perspective.

Previous investigations have shown that linemen produce greater peak power during vertical jump tasks over other positional groups.[5,7] Similar results were seen in the present study with linemen having significantly greater peak propulsive power than both the midskill and skill groups. The present investigation took this comparison a step further by exploring the two components used in the calculation of power (force and velocity) at the point of peak power. These data provide supporting evidence of the greater reliance on force by the lineman group which was significantly higher than both the other groups. This coincides with the skill group having greater velocity at peak propulsive power than both the lineman and midskill groups. In totality, this demonstrates differences in jumping strategies used between positional groups. This information can be of particular interest to practitioners when building athlete profiles and benchmarking athletes when performing a needs analysis before the initiation of training.

Merrigan et al.[5] reported that braking, deceleration, and propulsive force were greater in the lineman group compared to all other groups. However, in the present investigation, no differences were seen between groups in both peak and mean braking force with differences existing in propulsive peak and mean force. This can be potentially explained through differences in data analysis techniques. In the current investigation force values are reported as net force whereas Merrigan et al.[5] reported force as the absolute value. When using the absolute value differences can be a result of individuals having different body masses. This explains the greater force outputs for the lineman group across all force values reported previously where no differences were reported between groups within the braking phase or force at zero velocity in the current study. It is also important to note that differences in phase definition

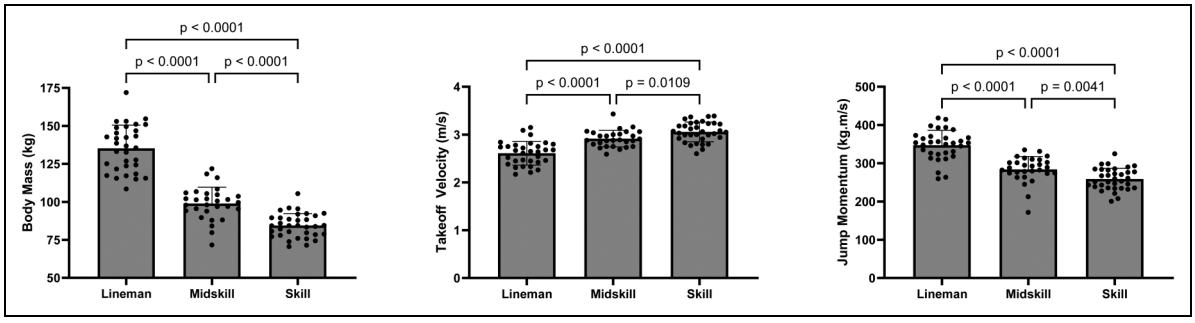

Figure 2. Mean and *SD* take-off momentum between positional groups.

can create confusion when comparing studies. The current study identified the braking phase as the point at which force crosses the individual's body mass after movement initiation. Though using the same braking phase term, Merrigan et al.[5] defined the braking phase as the point of minimum force to the point of peak eccentric velocity. Then the deceleration phase is defined as the end of the braking phase to the point at which an individual reaches zero velocity which corresponds to the braking phase in the current study. The difference in phase definition is important to note both for researchers and practitioners alike as it can create confusion when comparing results.

Additionally, it has been shown that differences in physical performance testing and specifically CMJ performance exist between levels of competition in American football.[10,11] Though the present study and Merrigan et al.[5] both used Division I football athletes, the current study was from the lower Group of 6 level, while Merrigan et al.[5] were from the higher Power 4 level. As mentioned previously, force values between studies are difficult to compare because of the use of absolute or net force. However, metrics such as jump height, time to take-off and RSImod can be compared. Jump heights were higher for all positional groups in the Merrigan et al.[5] investigation though body mass values for each positional group were similar between studies. Additionally, the time to take-off was lower for all groups. These differences would then also create a difference in RSImod between studies as well. Future investigations should explore differences between these two levels of NCAA Division I sports to aid in appropriate profiling and benchmarking of athletes. Another explanation for the differences between studies could potentially be a result of differences in the threshold used to identify movement onset. The current study employed the recommended use of the first force sample below 5SD of the weighing phase, while Merrigan et al.[5] used a 30N threshold. Previous investigations have shown differences in jump performance when using different movement onset thresholds.[18,27–29]

The current investigation also examined the take-off momentum as a potential metric to be used in American football athletes. As momentum generation is an important element of collision-based sports, monitoring the interplay of body mass and movement velocity in American football athletes could be of potential benefit. Similar to previous recommendations regarding the use of RSImod, when examining take-off momentum both components used in the calculation should be evaluated as well.[30] For example, if a lineman were to have a reduction in take-off momentum during the season it would be important both for strength and conditioning professionals as well as sports nutritionists to understand if this change was caused by a reduction in body mass or take-off velocity as this would be useful in creating an intervention. Similarly, when evaluating changes to offseason training, it would be important to understand if changes in take-off momentum were based on body mass changes or increases in take-off velocity.

Previous investigations involving rugby athletes have shown that positional differences existed in take-off momentum with forwards having greater take-off momentum than the backs, though the backs had reached greater jump heights.[14] The current study supports these earlier findings related to take-off momentum with the lineman group having a greater take-off momentum than both the midskill and skill groups. The additional body mass that the lineman possesses outweighs the slower movement velocity during the CMJ. It is also important to note the overlap between positional groups regarding take-off momentum (Figure 2). While group means were significantly different from each other, there were individuals at the bottom of the lineman group who displayed values similar to those in the skill group. Conversely, the top performers in the skill group generated take-off momentum similar to those in the lineman group. Thus, when profiling and benchmarking it is important to identify key performance indicators for each positional group. The individuals with the greatest take-off momentum within the lineman group also had the greatest body mass while in the skill group, the individuals with the greatest take-off momentum tended to have the greater movement velocities. This is worth further exploration as body mass has been identified as the leading contributing factor to both sprint and take-off momentum in previous investigations.[12,14,31]

This investigation is not without its limitations. This analysis was performed using a cross-sectional design of one team.

As noted there may be differences between levels of competition as previous investigations have had higher levels of force, greater jump heights, and shorter time to take-off. Thus, the current results may not be generalizable across all levels of football. This investigation was also performed at the beginning of the summer training period. As such the athletes had just returned from a short period of non-structured training. Thus, the data obtained may not have been indicative of the top performance each athlete could achieve. However, this was also the time in the training calendar when the greatest number of athletes were injury-free and factors such as the previous training block would not potentially influence the results.

Conclusion

The results of this investigation suggest that positional groups in American football utilize differing movement strategies during the CMJ. In totality, the findings of the current investigation can help aid practitioners in the design of targeted training programs by being able to better understand the positional differences across the sport of American football. Additionally, the use of take-off momentum may be of value when exploring CMJ performance in American football athletes. Take-off momentum could be of particular importance when practitioners are looking for metrics during athlete monitoring protocols that would have implications back to the sport itself. Lastly, the results of this study can be used in the talent identification process as well as in creating benchmarks and athlete profiles for strength and conditioning professionals and sport coaches alike.

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Data availability statement

The data that supports the findings of this study are available from the corresponding author, upon reasonable request.

IRB approval

Yes.

References

1. Gathercole R, Sporer B, Stellingwerff T, et al. Alternative countermovement-jump analysis to quantify acute neuromuscular fatigue. *Int J Sports Physiol Perform* 2015; 10: 84–92.
2. Taylor K-L, Chapman DW, Cronin JB, et al. Fatigue monitoring in high performance sport: a survey of current trends. *J Aust Strength Cond* 2012; 20: 12–23.
3. Donahue PT, Peel SA, Rush M, et al. Examining countermovement jump strategies between women's NCAA division I sports. *J Strength Cond Res* 2023; 37: 2052–2057.
4. McMahon JJ, Jones PA and Comfort P. Comparison of countermovement jump–derived reactive strength index modified and underpinning force-time variables between super league and championship rugby league players. *J Strength Cond Res* 2022; 36(1): 226–231.
5. Merrigan JJ, Rentz LE, Hornsby WG, et al. Comparisons of countermovement jump force- time characteristics among national collegiate athletic association division I American football athletes : use of principal component analysis. *J Strength Cond Res* 2021; 36: 411–419.
6. Bayliff GE, Jacobson BH, Moghaddam M, et al. Global positioning system monitoring of selected physical demands of NCAA division I football players during games. *J Strength Cond Res* 2019; 33: 1185.
7. Lockie RG, Lazar A, Orjalo AJ, et al. Profiling of junior college football players and differences between position groups. *Sports* 2016; 4: 41.
8. Robbins DW and Young WB. Positional relationships between various sprint and jump abilities in elite American football players. *J Strength Cond Res* 2012; 26: 388.
9. Sanchez E, Weiss L, Williams T, et al. Positional movement demands during NFL football games: a 3-year review. *Appl Sci* 2023; 13: 9278.
10. Fry AC and Kraemer WJ. Physical performance characteristics of American collegiate football players. *J Strength Cond Res* 1991; 5: 126.
11. Garstecki MA, Latin RW and Cuppett MM. Comparison of selected physical fitness and performance variables between NCAA division I and II football players. *J Strength Cond Res* 2004; 18: 292–297.
12. McMahon JJ, Lake JP, Ripley NJ, et al. Vertical jump testing in rugby league: a rationale for calculating take-off momentum. *J Appl Biomech* 2020; 36: 370–374.
13. Jalilvand F, Banoocy NK, Rumpf MC, et al. Relationship between body mass, peak power, and power-to-body mass ratio on sprint velocity and momentum in high-school football players. *J Strength Cond Res* 2019; 33: 1871.
14. McMahon JJ, Lake JP and Comfort P. Identifying and reporting position-specific countermovement jump outcome and phase characteristics within rugby league. *PLoS One* 2022; 17: e0265999.
15. Carbuhn AF, Womack JW, Green JS, et al. Performance and blood pressure characteristics of first-year national collegiate

athletic association division I football players. *J Strength Cond Res* 2008; 22: 1347.
16. Chavda S, Bromley T, Jarvis P, et al. Force-time characteristics of the countermovement jump: analyzing the curve in excel. *Strength Cond J* 2018; 20: 67–77.
17. McMahon JJ, Suchomel TJ, Lake JP, et al. Understanding the key phases of the countermovement jump force-time curve. *Strength Cond J* 2018; 40: 96–106.
18. Donahue PT, Hill CM, Wilson SJ, et al. Squat jump movement onset thresholds influence on kinetics and kinematics. *Int J Kinesiol Sports Sci* 2021; 9: 1–7.
19. Harry JR, Barker LA, Tinsley GM, et al. Relationships among countermovement vertical jump performance metrics, strategy variables, and inter-limb asymmetry in females. *Sports Biomech* 2024; 23: 1009–1027.
20. Lake JP, Mundy PD, Comfort P, et al. Effect of barbell load on vertical jump landing force-time characteristics. *J Strength Cond Res* 2021; 35: 25.
21. Owen NJ, Watkins J, Kilduff LP, et al. Development of a criterion method to determine peak mechanical power output in a countermovement. *J Strength Cond Res* 2014; 28: 1552–1558.
22. Ebben WP and Petushek EJ. Using the reactive strength index modified to evaluate plyometric performance. *J Strength Cond Res* 2010; 24: 1983–1987.
23. Moir GL. Three different methods of calculating vertical jump height from force platform data in men and women. *Meas Phys Educ Exerc Sci* 2008; 12: 207–218.
24. Koo TK and Li MY. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J Chiropr Med* 2016; 15: 155–163.
25. Cormack SJ, Newton RU, McGuigan MR, et al. Reliability of measures obtained during single and repeated countermovement jumps. *Int J Sports Physiol Perform* 2008; 3: 131–144.
26. Johnson QR, Yang Y, Cabarkapa D, et al. Key performance indicators for college American football starters: an exploratory study. *J Funct Morphol Kinesiol* 2025; 10: 19.
27. Pérez-Castilla A, Fernandes JFT, Rojas FJ, et al. Reliability and magnitude of countermovement jump performance variables: influence of the take-off threshold. *Meas Phys Educ Exerc Sci* 2021; 25: 227–235.
28. Pérez-Castilla A, Rojas FJ and García-Ramos A. Assessment of unloaded and loaded squat jump performance with a force platform: which jump starting threshold provides more reliable outcomes? *J Biomech* 2019; 92: 19–28.
29. Pérez-Castilla A, Rojas FJ and García-Ramos A. Reliability and magnitude of loaded countermovement jump performance variables: a technical examination of the jump threshold initiation. *Sports Biomech* 2022; 21: 622–636.
30. Bishop C, Turner A, Jordan M, et al. A framework to guide practitioners for selecting metrics during the countermovement and drop jump tests. *Strength Cond J* 2021; 44: 95–103.
31. McErlain-Naylor SA and Beato M. Factors influencing the jump momentum – sprint momentum correlation: a data simulation. *Eur J Sport Sci* 2022; 22: 1847–1855.